



GaN-on-SiC RF: Poised for Rapid Adoption

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Wolfspeed Campus



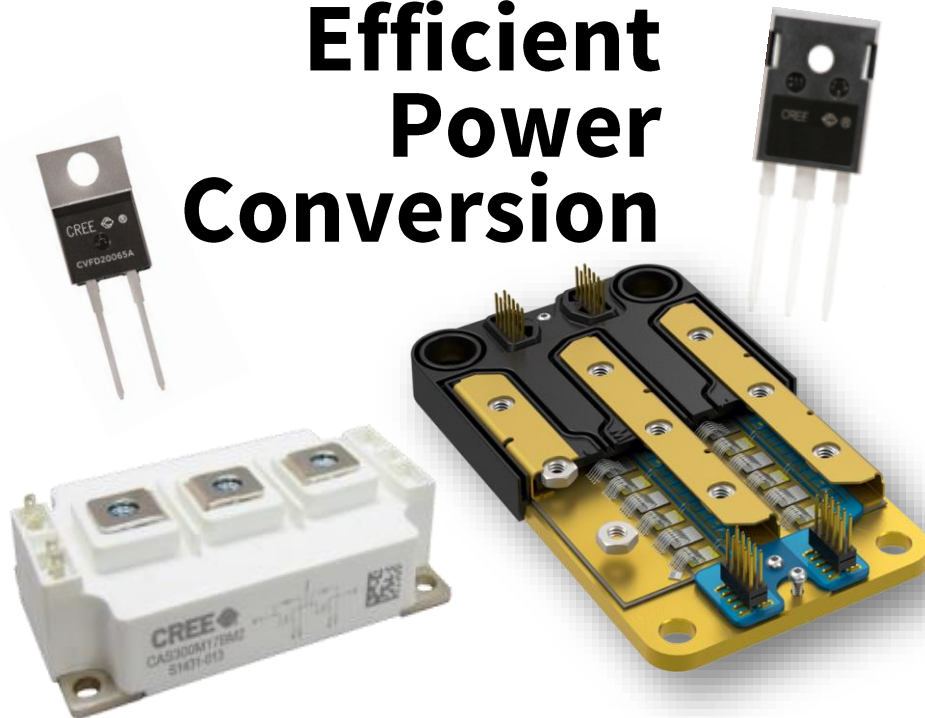
- HQ in Research Triangle Park, North Carolina, USA
- ~\$175 million in revenue (FY2016)
- ~550 employees;
 - 20% have advanced degrees
- 1,256 patents worldwide, with 592 patent applications pending
- 46,000 square foot fabrication facility on 55 acres
- Class 10, 100, and 1000 Clean Rooms

WOLFSPEED: MARKET'S BROADEST, MOST FIELD-TESTED SiC POWER AND RF GaN PORTFOLIO

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Leading innovator and manufacturer
of devices for

Efficient Power Conversion



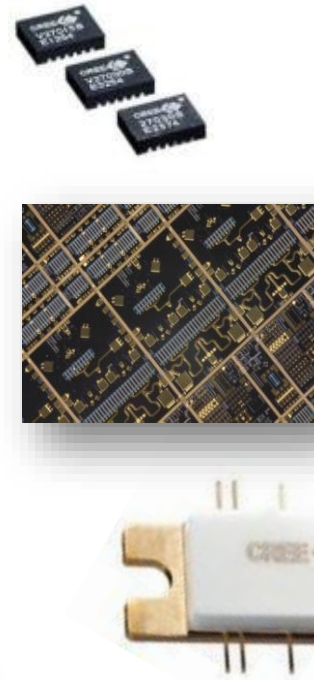
MOSFETs: 900-1700V, $R_{DS(on)}$: 25m Ω -1 Ω , Die and packaged.

Diodes: 600-1700V, Die and packaged.

Modules: 45mm, 1200V, 10-20kW & 62mm 1200V, 1700V,
40-300kW devices with accompanying gate drivers.

A leading supplier of

GaN RF devices for wireless communications



Devices with a frequency range from **DC to 18GHz**.

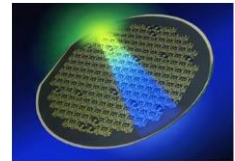
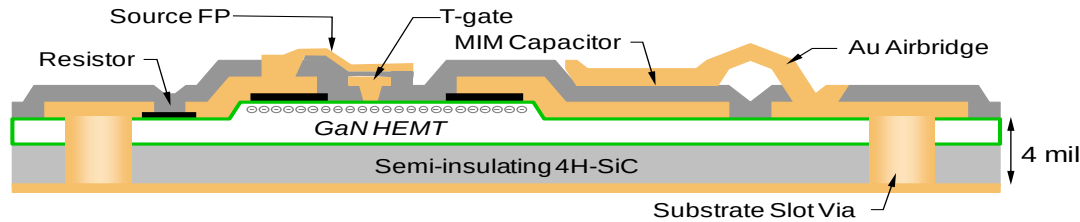
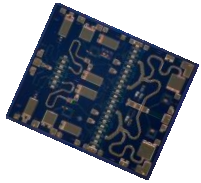
GaN HEMT transistors in die and packaged form (pre or fully-matched for band specific applications or unmatched for broadband applications).

GaN HEMT MMIC amplifiers in die and packaged form.

The **foundry** supports custom GaN HEMT MMICs, including high power amplifiers, LNA, switches and phase shifters.

CREE GaN MMIC FOUNDRY PROCESSES

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■ G28V3 C-Band MMIC Process

- Broad Band, Moderate Power
- 28 V bias, 120V Breakdown
- 0.4 μm Gate length
- 4 W/mm
- DC-8 GHz Performance
- Status: Released since 2007

■ G28V4 X/K_U GaN MMIC process

- Wide Band, Moderate Power
- 28 V bias, 120V Breakdown
- 0.25 μm Gate length
- 4 W/mm
- DC-18 GHz Performance
- Status: Released April 2012

■ G50V3 C-Band MMIC process

- Broad Band, High Power
- 50 V bias, 150V Breakdown
- 0.40 μm Gate length
- 7 W/mm
- DC-8 GHz Performance
- Status: Released May 2012

■ G40V4 X/K_U MMIC process

- High Frequency, High Power
- 40 V bias, 120V Breakdown
- 0.25 μm Gate length
- 6 W/mm
- DC-18 GHz Performance
- Status: Released April 2012

FIELD FAILURE RATE

- Wolfspeed's GaN devices have been shipping since 2007
- The documented field failure rate is very low and on par with more established RF technologies
- This result demonstrates the reliability of Wolfspeed GaN

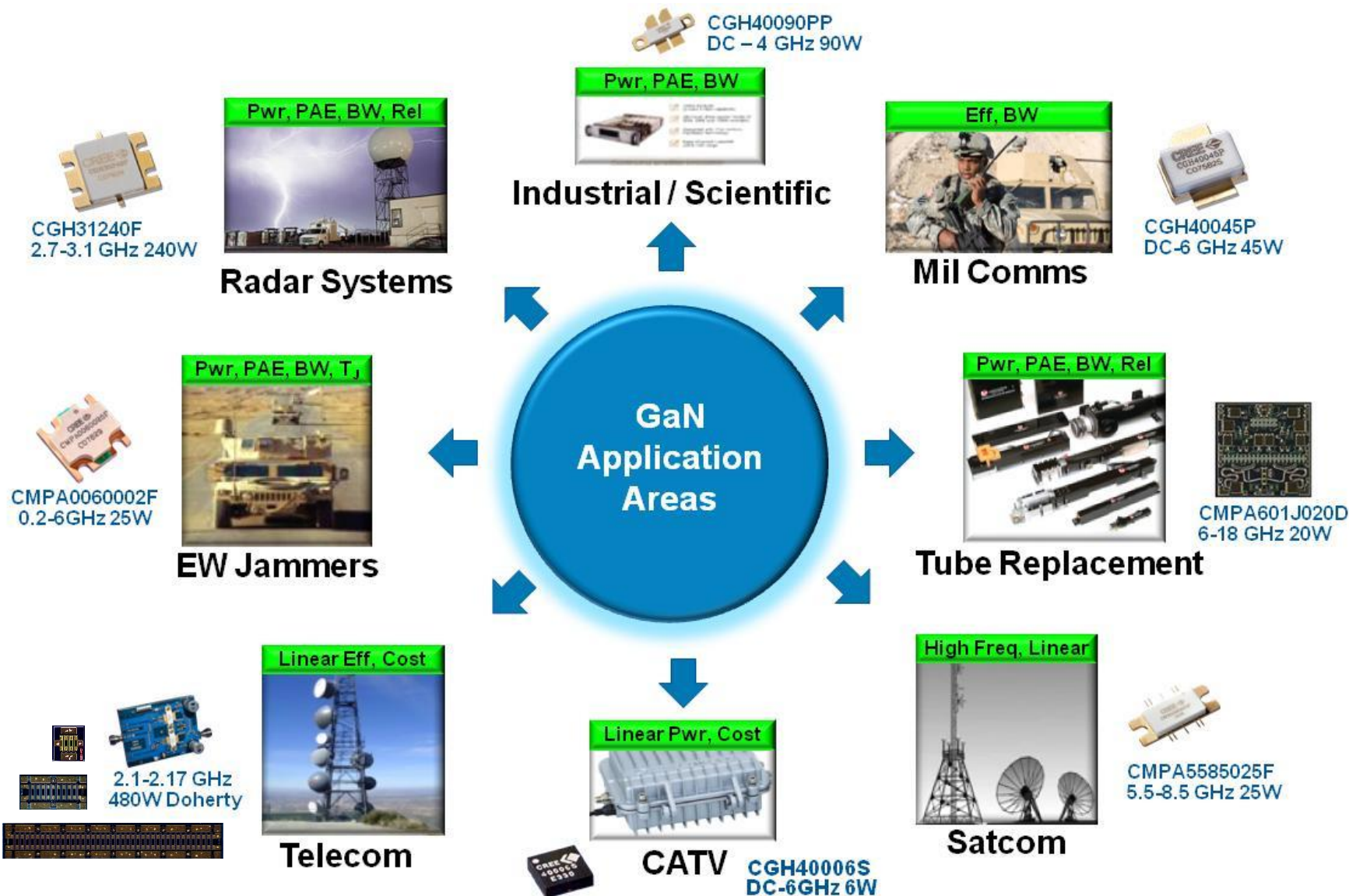
Technology	Fielded Device Hours (billions)*	FIT Rate (valid field fails per billion device hours)**
G28V3	78.7	2.1
G50V3	25.9	0.2
G40V4	2.2	1.9

*Calculated as: today's date minus confirmed ship date minus 90 days (allowing for time to put into service) * 12 hours per day

**Calculated as: 2 times the number of valid field failures (excludes engineering evaluations, as-received visual defect escapes or issues, as-received test escapes, packaging and assembly quality issues, and other extraneous non-Wolfspeed related issues) divided by fielded device hours; includes an additional factor for statistical confidence margin

GaN PROCESS SUPPORTS MULTIPLE APPLICATION AREAS

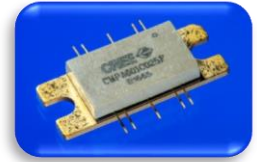
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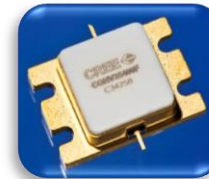
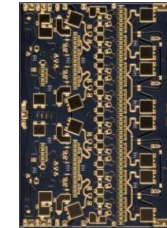
WOLFSPEED RF PRODUCT FAMILY

- Portfolio has about 90 products
- Uses all four released processes
 - G28V3, G50V3, G28V4, G40V4
 - 0.4um and 0.25um

Packaged MMIC



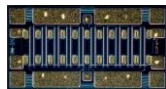
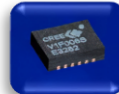
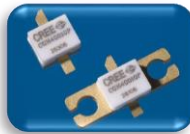
MMIC



Fully Matched Transistors



Unmatched Transistors



Bare Die



Increasing value to end customer in terms of ease of use and performance

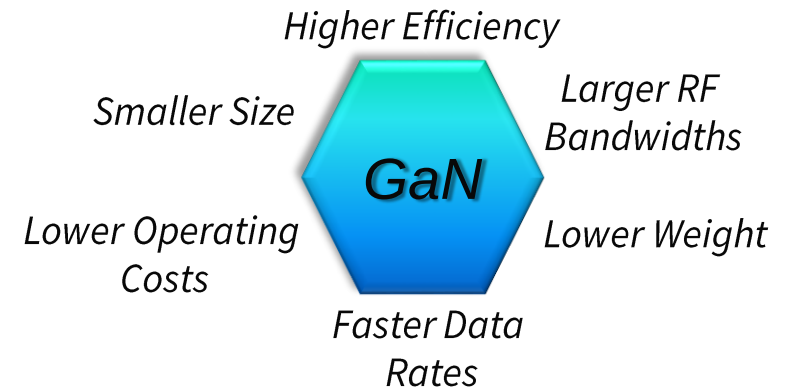
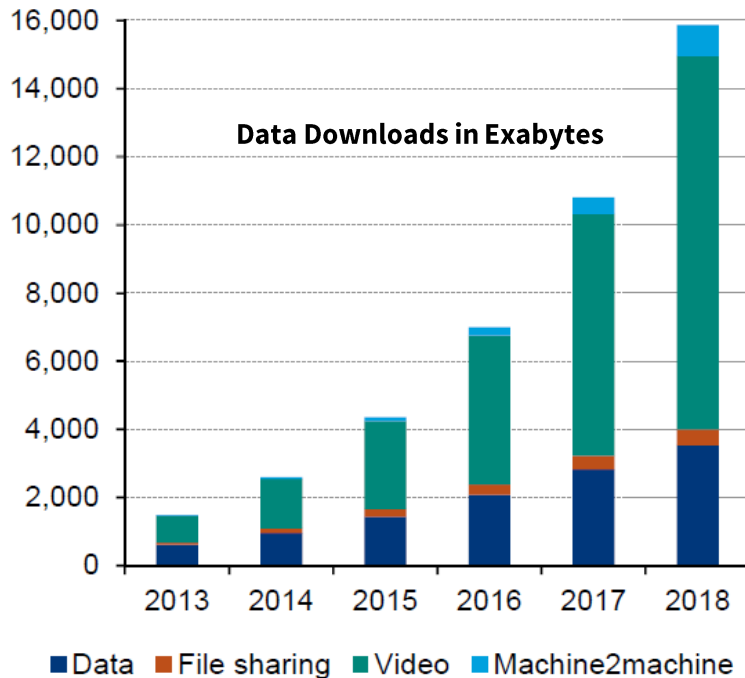


MARKET TRENDS

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WHY GaN FOR RF: DISRUPTIVE TECHNOLOGY

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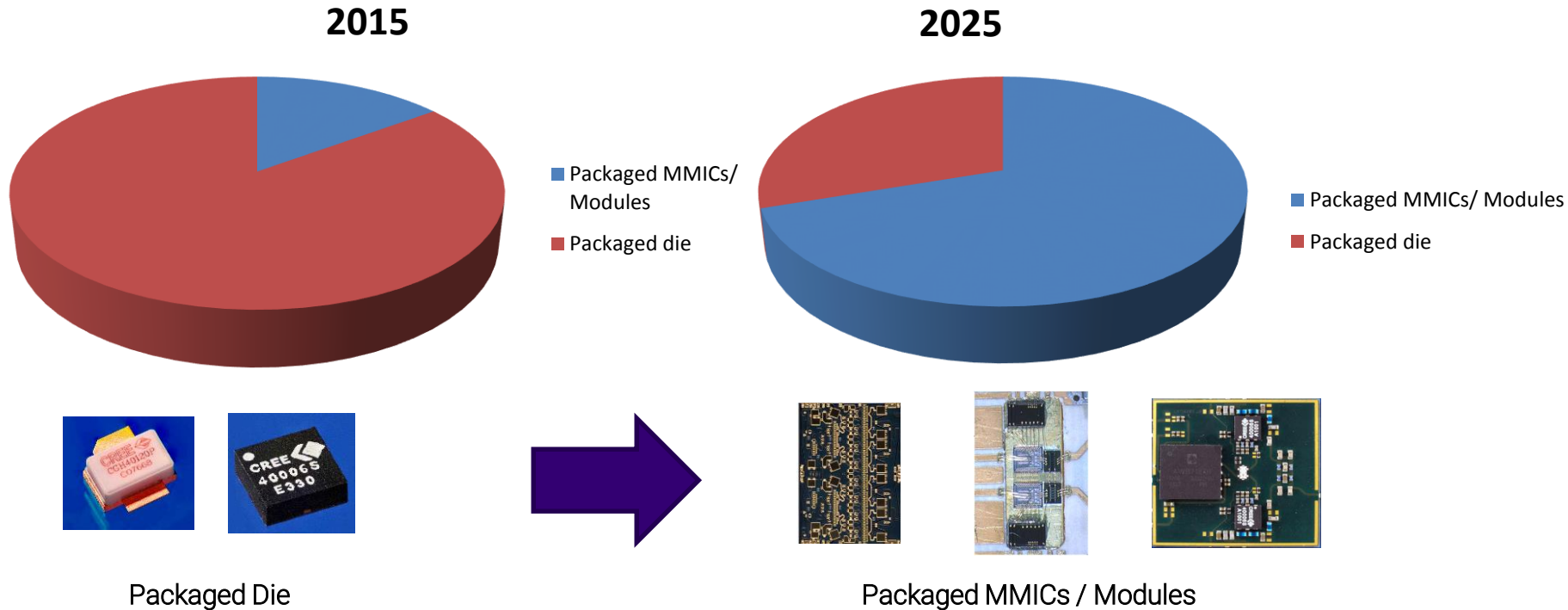


GaN provides game changing system solutions

- Higher data rates range
- Higher efficiency
- Lighter weight
- Improved target detection
- Lower CapEx, OpEx
- Smaller

RF POWER COMPONENT INTEGRATION TRENDS

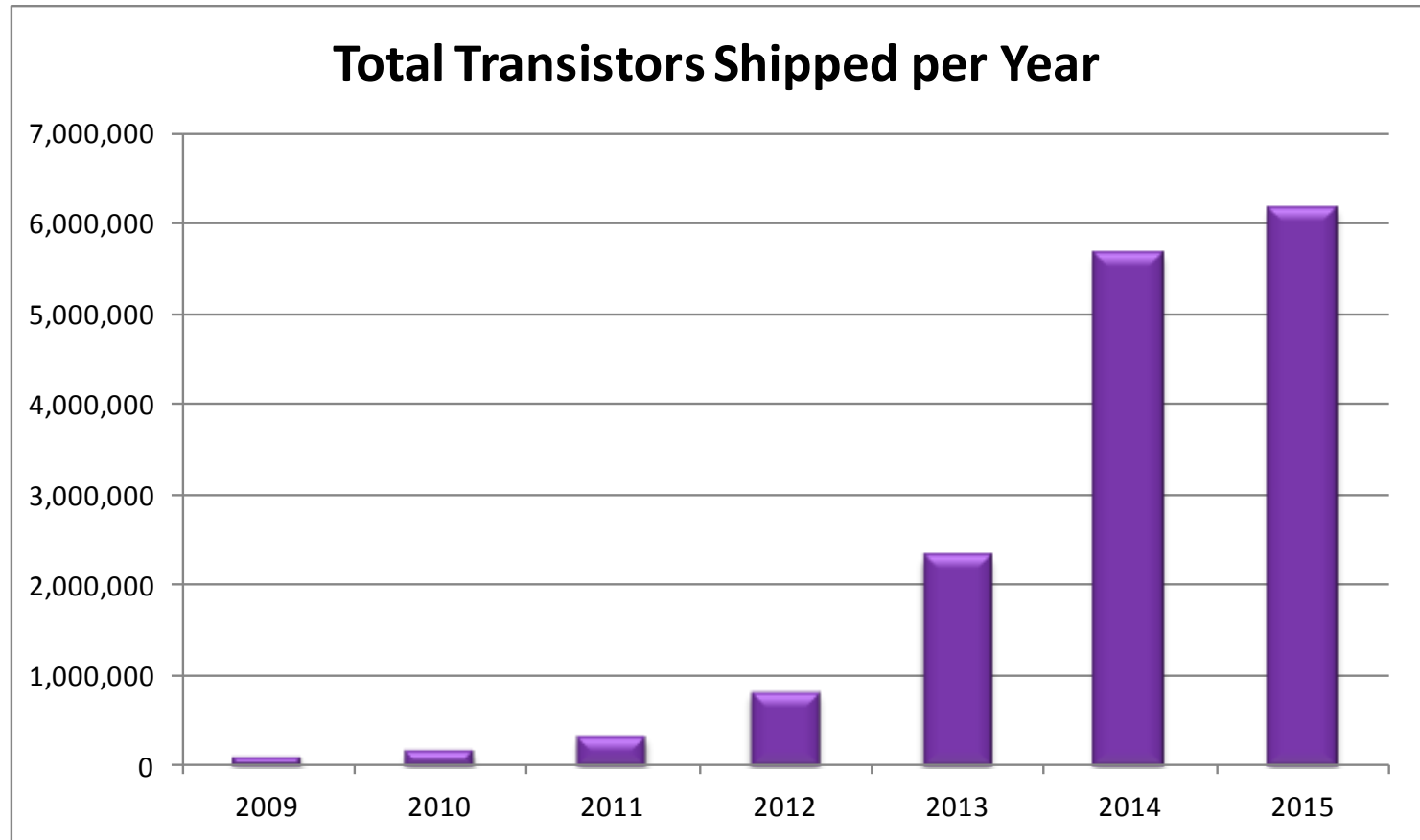
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- Higher levels of integration will continue to be the trend
- Packaged die will progress to packaged modules and MMICs depending on power level, frequency and application
 - Lower power/higher frequency → MMICs (RFICs)
 - Higher power / lower frequency → Modules

TOTAL GaN TRANSISTORS SHIPPED PER CALENDAR YEAR

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SUMMARY OF MARKET TRENDS DRIVING ADOPTION

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- Insatiable need for data continues to grow wireless network needs
 - 4G Advanced LTE
- Continued migration of military radar, EW, and other communications to GaN
 - Replacement of RF tubes with GaN SSPAs
(military, satcom, commercial, consumer)
 - Higher power capability at high frequencies

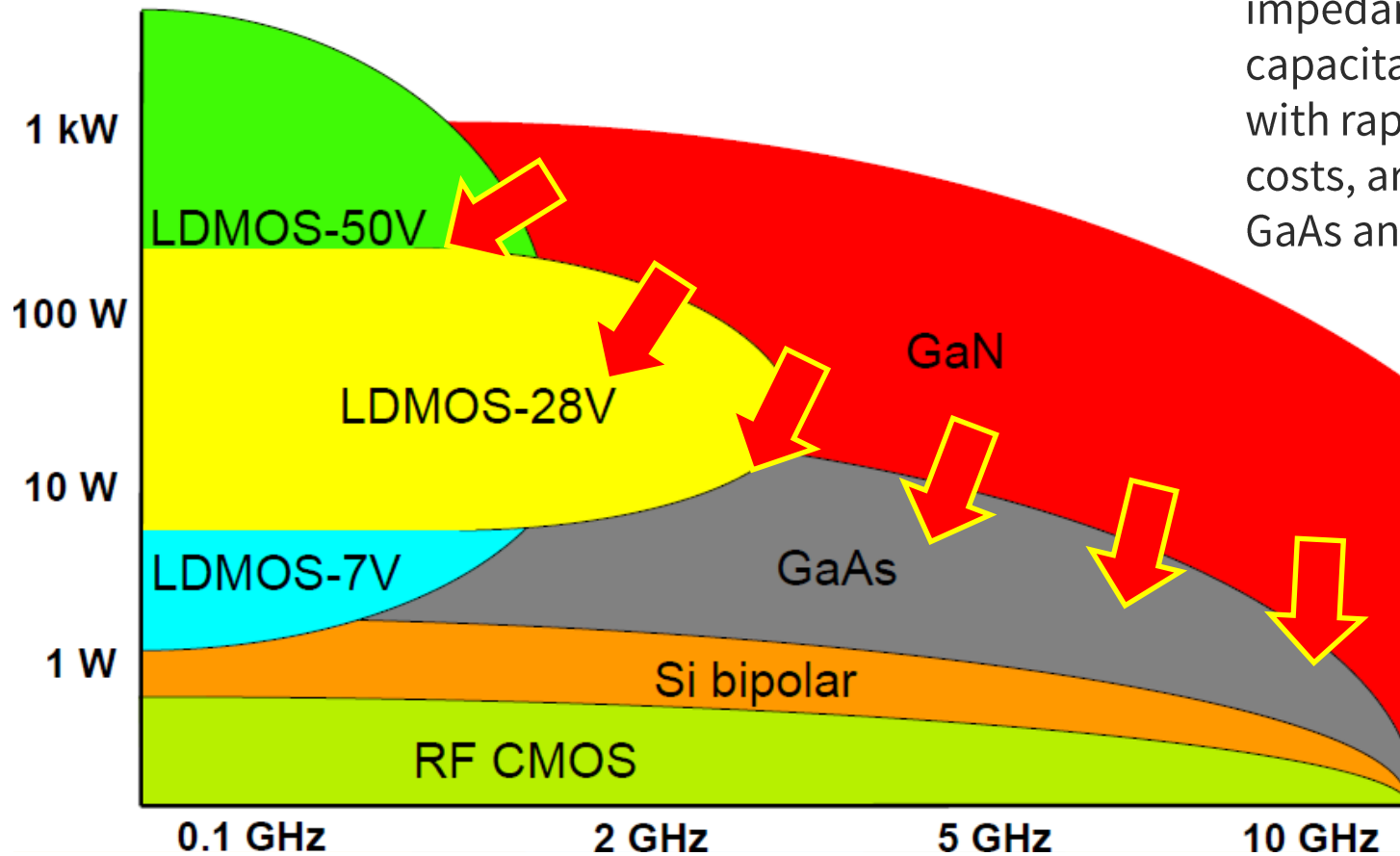


COMMERCIAL ADOPTION IN CELLULAR INFRASTRUCTURE

TM

WHY GaN FINDS ITS APPLICATIONS SPACE

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Higher gain, power density, higher impedances and lower capacitances of GaN, with rapidly decreasing costs, are pushing into GaAs and Si markets

WHY USE GaN IN COMMERCIAL TELECOMMUNICATIONS ?

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1. Enabling Benefits (Game Changer)

- Tough to accomplish any other way
(i.e. > 100 MHz channel bandwidths)

2. Feature Enhancements

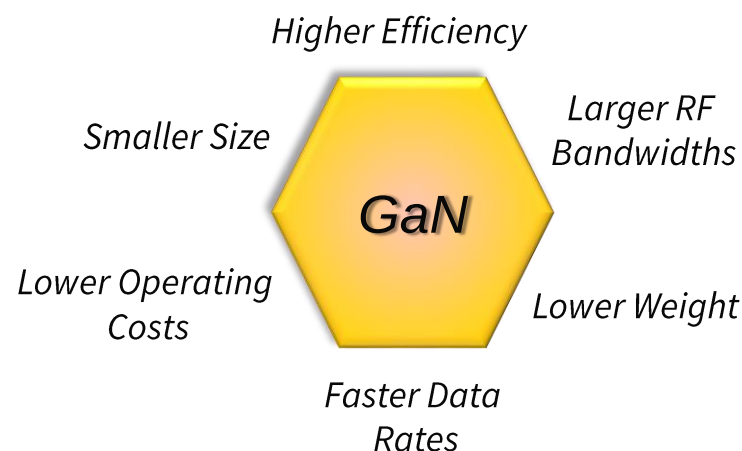
- Provides significant performance gain
(e.g. improved linear efficiency)
- Higher power for extended range

3. Lower Acquisition Cost / CAPEX Savings

- Overall Bill of material savings

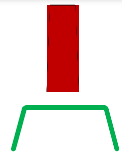
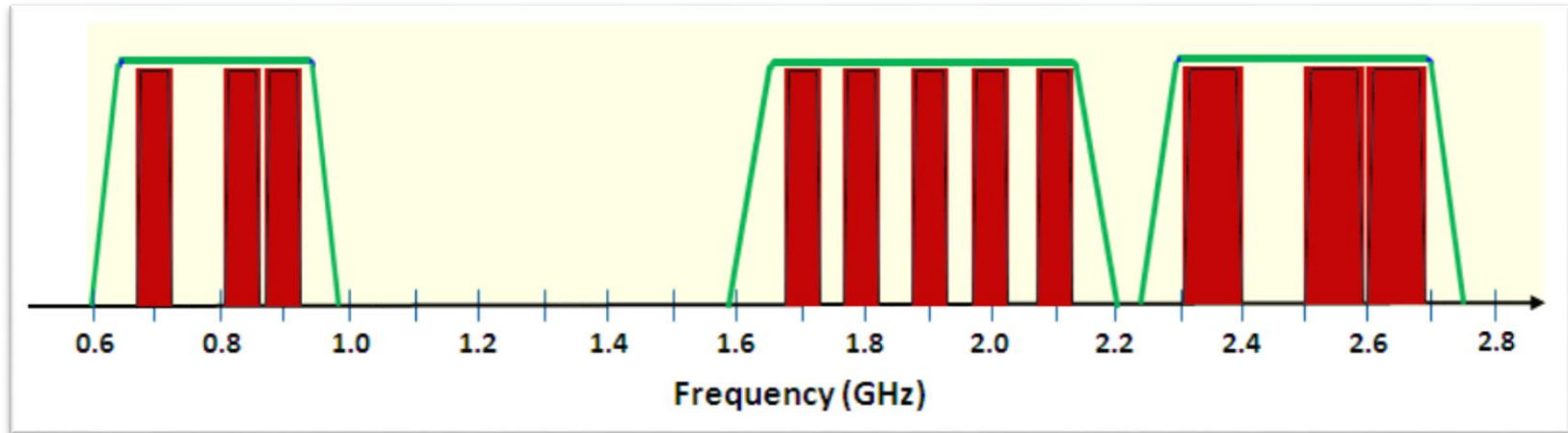
4. Life Cycle / OPEX Savings

- Operating cost savings
(e.g. less maintenance and lower power consumption)



GaN'S BANDWIDTH ADVANTAGE REDUCES OPERATOR INVESTMENT

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- Common LTE Telecom Bands (one silicon amplifier per band)

- Multiple bands covered by single GaN amplifier

- GaN covers multiple bands with a single amplifier
 - reduces operator capex
 - simplifies inventory management
- GaN's advantage increases as frequency and bandwidth increases

WHY IS GaN REPLACING LDMOS IN DOHERTY AMPLIFIERS?

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Property	Si	4H-SiC	GaN
Band Gap (eV)	1.12	3.26	3.4
Breakdown Field (10 ⁶ V/cm)	0.25	2.2	3
Power Density (W/mm)	1.5	-	8.0
Thermal Conductivity (W/Kcm)	1.5	4.9	1.5
Ft (GHz)	~ 5.0	-	15

Feature

Efficiency
Bandwidth (data rate)
RF Bandwidth
Power consumption
Size
Cost
Weight

Typical GaN Benefit

> 15% improvement in amplifier efficiency
100% wider channel bandwidths
Carrier aggregation
> 25% lower power consumption
> 50% reduction in PA module size
15% typical overall BOM cost reduction
Reduced weight / heat sink requirements





COMMERCIAL ADOPTION IN RADAR AND SATELLITE COMMUNICATIONS

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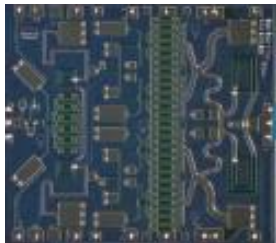
GaN MMICs POISED TO SATISFY NEXT GENERATION MILITARY SYSTEMS



Shipboard Defense Radar and Doppler Weather Radar



10 kW S-Band
Phase Combined
Amplifier System



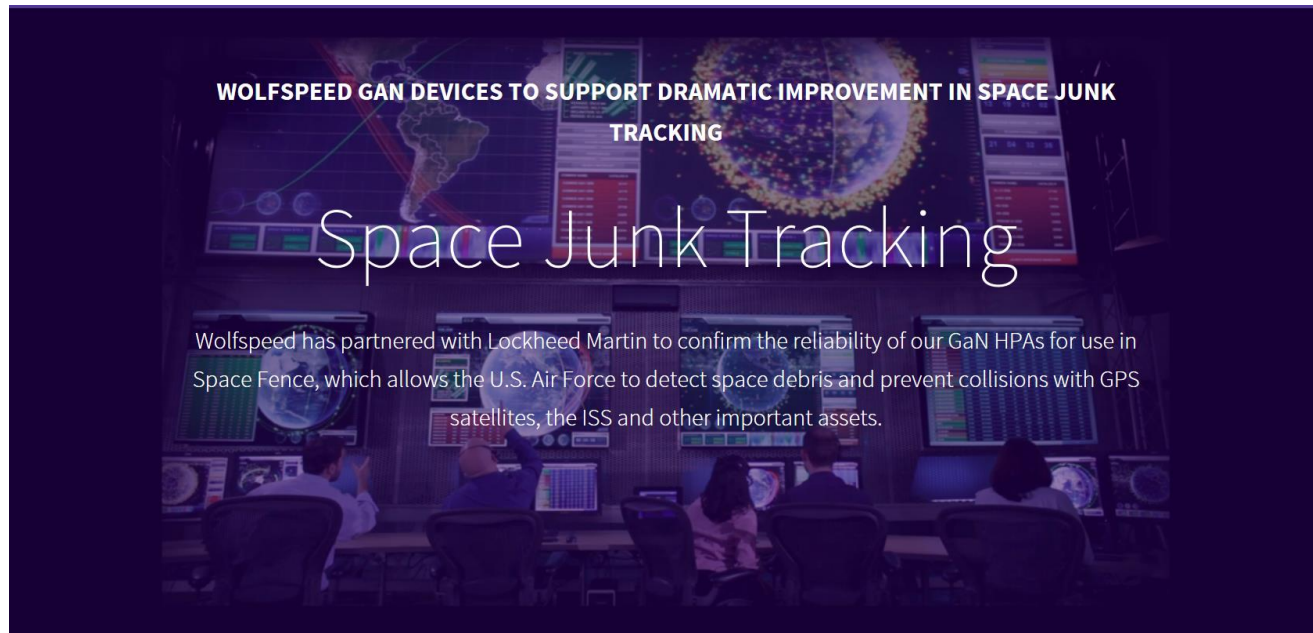
75 W RF GaN MMIC Die

75 W RF GaN MMIC



800 W S-Band Power
Amplifier Module

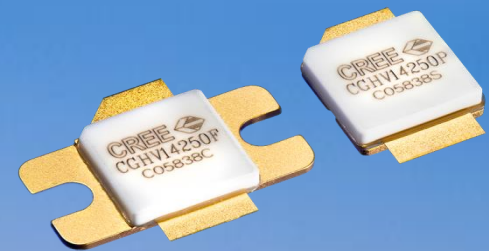
SPACE FENCE



- Wolfspeed partnered with Lockheed Martin to provide GaN high power amplifiers for the U.S. Air Force's Space Fence
- Improves timeliness of detection of space events that could threaten GPS satellites or the International Space Station
- Located in the Marshall Islands Space Fence will accurately track an estimated 500,000 objects (10x compared to current system)



50V RF COMPONENTS FOR L-BAND RADAR



50V GaN HEMT
250W & 500W Solutions

GaN MMICs POISED TO SATISFY NEXT GENERATION COMMUNICATIONS SYSTEMS



Marine Communications, Satellite Communications, High Definition Data Links



60 W RF GaN MMIC Die



100 W RF GaN Transistor



200 W Ku-Band Block Upconverter



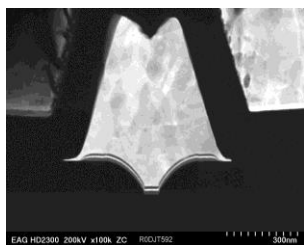
NEW AREAS FOR ADOPTION

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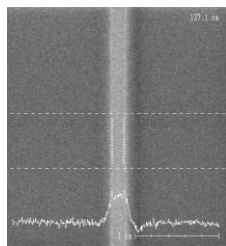
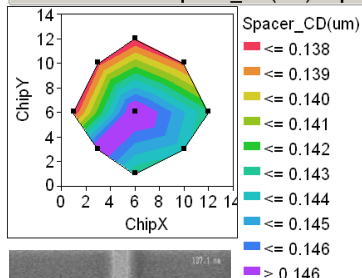
FUTURE MARKET NEEDS THAT WILL DRIVE FURTHER ADOPTION

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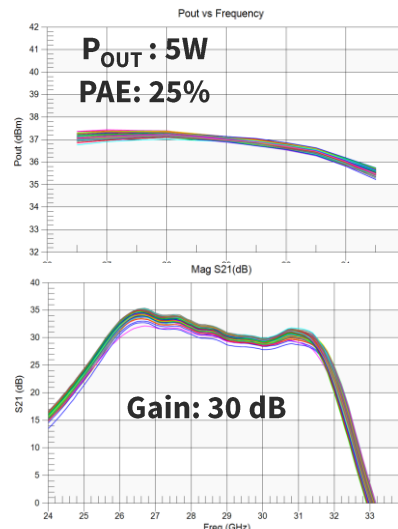
- Insatiable need for data will continue to grow wireless network needs (IoT)
 - Infrastructure (4G Advanced LTE migrating to 5G)
 - Requiring higher frequencies in C, X, Ku and Ka-bands
 - WiFi (greater demands on outdoor access points)
 - M2M (white goods, automotive traffic management/collision avoidance)
 - Expansion of commercial satellite networks
- Higher levels of integration (MMICs, Modules, Mixed mode RF/Digital)
 - envelope tracking, integrated PA/Modulator
- Continued migration of military radar, EW, communications to GaN
- GaN Migration from high to lower power applications access points, routers, terminals



Contour Plot for Spacer_CD(um) ExposeTime=250ms



V5 0.14 μm Optical Spacer Process



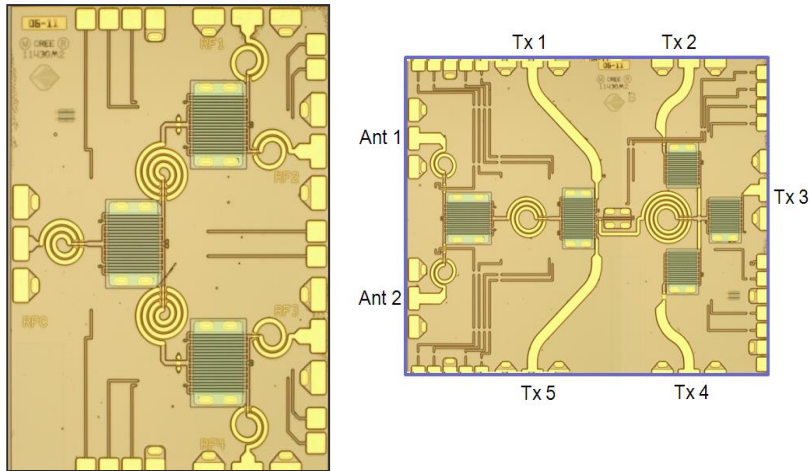
3-Stage 5W Ku Band PA (27.5 – 30.5 GHz)



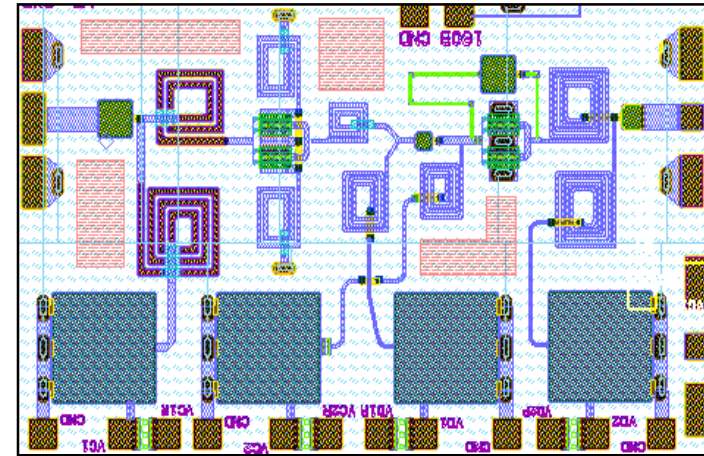
Parameter	Goal	3.5 GHz	39 GHz
Device		2x360 μm	6x50 μm
Pout	3W/mm@35 GHz	5.6 W/mm	3.3 W/mm
PAE	35% @ 35 GHz	62%	33%
Gain _{Assoc}	8 dB @ 35 GHz	14.2 dB	9 dB
Gain _{SS}	11 dB @ 35 GHz	19.1 dB	11 dB
Vd _q	28V	28 V	28 V
Id _q	100 mA/mm	100 mA/mm	100 mA/mm

V5 Load Pull Performance vs. Goals

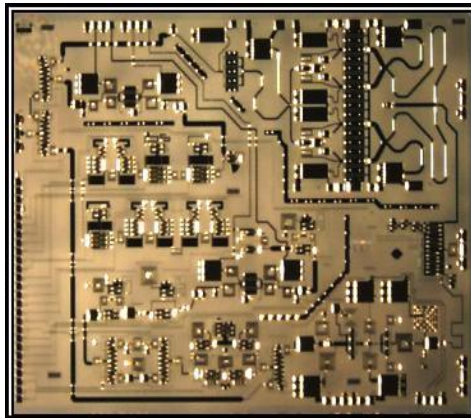
- V5 mm-wave Process in Qualification
- Utilizes Smaller Vias and 3 mil Substrates
- Currently Released to “Early Adopters”
- Excellent for Ku Band Satcom / Mil Comms
- Higher Gain Alternative to 28V V4



T/R and Multi-Pole Multi-Throw Switches



LNAs



Multi-Function Circuits

- **High Power Switches**
- **High TOI LNAs**
- **Limiters, Mixers**
- **Multi-Function Phase Gain / Control**
- **Class E combined with Dual Gate for Higher Efficiency, Variable Power**



*Wolfsp*eed™

Leading the Pack™